CSE 242
Homework # 1

October 9, 2026

Erik Chao
etchao@ucsc.edu

Part 1
Q1: Given matrices:

$$A = \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}$$

Compute:

$$AB, \quad BA, \quad (A+B)^T, \quad AB^T, \quad \text{trace}(A), \quad \text{trace}(AB).$$

Ans:

$$A = \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}$$

$$\begin{aligned} AB &= \begin{bmatrix} (3)(1)+(4)(0) & (3)(2)+(4)(-1) \\ (2)(1)+(1)(0) & (2)(2)+(1)(-1) \end{bmatrix} \\ &= \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} BA &= \begin{bmatrix} (1)(3)+(2)(2) & (1)(4)+(2)(1) \\ (0)(3)+(-1)(2) & (0)(4)+(-1)(1) \end{bmatrix} \\ &= \begin{bmatrix} 7 & 6 \\ -2 & -1 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} (A+B)^T &= \left(\begin{bmatrix} 4 & 6 \\ 2 & 0 \end{bmatrix} \right)^T \\ &= \begin{bmatrix} 4 & 2 \\ 6 & 0 \end{bmatrix} \end{aligned}$$

$$AB^T = \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix}$$
$$= \begin{bmatrix} 11 & -4 \\ 4 & -1 \end{bmatrix}$$

$$\text{trace}(A) = 3 + 1$$
$$= 4$$

$$\text{trace}(AB) = 3 + 3$$
$$= 6$$

1. Q2: For the following matrix:

$$M = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

Calculate the eigenvalues and eigenvectors. Verify your solution by showing $Mv = \lambda v$.

Ans: Using determinant to find eigenvalues: $det(M - \lambda I) = 0$:

$$det([[2, -1], [-1, 2]] - \lambda[[1, 0], [0, 1]]) = 0$$
$$det([[2 - \lambda, -1], [-1, 2 - \lambda]]) = 0$$
$$(2 - \lambda)(2 - \lambda) - (-1)(-1) = 0$$
$$4 + \lambda^2 + \lambda - 1 = 0$$
$$\lambda^2 + \lambda + 3 = 0$$
$$\lambda_1 = 3, \lambda_2 = 1$$

Using the eigenvalues, we can find the eigenvectors to prove that $Mv = \lambda v$:

$$([[2 - \lambda_1, -1], [-1, 2 - \lambda_1]])$$
$$([[2 - 3, -1], [-1, 2 - 3]])$$
$$= ([[-1, -1], [-1, -1]][[x], [y]] = [[0], [0]]$$

Let $v_1 = [[1], [-1]]$. We can verify $Mv_1 = \lambda_1 v_1$

$$\begin{aligned} Mv_1 &= [[2, -1], [-1, 2]] \cdot [[1], [1]] = [[3], [-3]] \\ &= 3[[1], [-1]] \\ &= \lambda_1 v_1 \end{aligned}$$

To solve for λ_2:

$$\begin{aligned} &([[2 - \lambda_2, -1], [-1, 2 - \lambda_2]]) \\ &([[2 - 1, -1], [-1, 2 - 1]]) \\ &= [[1, -1], [-1, 1]] \cdot [[x], [y]] = [[0], [0]] \end{aligned}$$

Let $v_2 = [[1], [1]]$. We can verify $Mv_2 = \lambda_2 v_2$:

$$\begin{aligned} Mv_2 &= [[2, -1], [-1, 2]] \cdot [[1], [1]] = [[3], [1]] \\ &= 1[[1], [1]] \\ &= \lambda_2 v_2 \end{aligned}$$

$\therefore Mv = \lambda v$

1. Q3: Prove the following property of the trace operator:

$$\text{trace}(AB) = \text{trace}(BA)$$

Ans: $trace(AB) = 6$ (Obtained from Q1)
$trace(BA) = [[7, 6], [-2, -1]] = 6$
$\therefore trace(AB) = trace(BA)$

1. Show step-by-step that the inverse of the matrix

$$C = \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}$$

exists, and compute its inverse.

Ans:

$$C = \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}$$

For a 2×2 matrix,

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

the inverse is

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Therefore,

$$\begin{aligned} C^{-1} &= \frac{1}{(4)(6) - (7)(2)} \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix} \\ &= \frac{1}{24 - 14} \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix} \\ &= \frac{1}{10} \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix} \\ &= \begin{bmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{bmatrix}. \end{aligned}$$

The given matrix follows the properties of an inverse matrix: 1. It is a square

2. It is non-singular

3. Let A be any matrices, in which $AA^{-1} = I$, where I is the identity matrix.

4. $A^{-1}A = I$

To prove that $AA^{-1} = I$:

$$\begin{aligned} CC^{-1} &= \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{bmatrix} \\ &= \begin{bmatrix} (4)(0.6) + (7)(-0.2) & (4)(-0.7) + (7)(0.4) \\ (2)(0.6) + (6)(-0.2) & (2)(-0.7) + (6)(0.4) \end{bmatrix} \\ &= \begin{bmatrix} 2.4 - 1.4 & -2.8 + 2.8 \\ 1.2 - 1.2 & -1.4 + 2.4 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I. \end{aligned}$$

To prove that $A^{-1}A = I$:

$$\begin{aligned} C^{-1}C &= \begin{bmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{bmatrix} \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix} \\ &= \begin{bmatrix} (0.6)(4) + (-0.7)(2) & (0.6)(7) + (-0.7)(6) \\ (-0.2)(4) + (0.4)(2) & (-0.2)(7) + (0.4)(6) \end{bmatrix} \\ &= \begin{bmatrix} 2.4 - 1.4 & 4.2 - 4.2 \\ -0.8 + 0.8 & -1.4 + 2.4 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I. \end{aligned}$$

$\therefore$ The inverse of C is

$$\boxed{C^{-1} = \begin{bmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{bmatrix}}$$

which is a valid inverse of C.

Part 2

1. Q1: Suppose a random variable X follows a discrete distribution:

$$P(X = k) = c \cdot \tfrac{3^k}{k!}, k = 0, 1, 2, ...$$

(a) Find the normalization constant c.
(b) Compute the expectation $E[X]$ and variance $Var(X)$

Ans:
(a) To find the normalization constant c, we will use the total probability rule, where:

$$\sum P(X = k) = 1$$

Thus, we can substitute $P(X = k)$ from the original equation with $\sum$:

$$\sum_{k=0}^{\infty} c \cdot \tfrac{3^k}{k!} = 1$$

We can factor out c since it does not depend on k:

$$c \sum_{k=0}^{\infty} \frac{3^k}{k!} = 1$$

Using the Taylor series expansion:

$$e^x = c \sum_{k=0}^{\infty} \frac{3^k}{k!}$$

We can then use $x = 3$, where

$$c \sum_{k=0}^{\infty} \frac{3^k}{k!} = e^3$$

Now, to substitute back the law of total probability rule and solve for c:

$$\begin{gathered} c \cdot e^3 = 1 \\ c = e^{-3} \\ c = \frac{1}{e^3} \end{gathered}$$

$\therefore$ the constant c is $\frac{1}{e^3}$.

(b) To compute the expectation and variance, we must solve for the expectation first. We will use the definition of expectation for a discrete random variable:

$$E[X] = \sum_{k=0}^{\infty} k \cdot P(X = k)$$

We can substitute $P(X = k) = e^{-3} \frac{3^k}{k!}$, as well as start from $k = 1$, since $k = 0$ means the total summation would be 0:

$$E[X] = \sum_{k=1}^{\infty} k \cdot e^{-3} \frac{3^k}{k!}$$

After, since $k/k! = 1/(k-1)!$:

$$E[X] = e^{-3} \sum_{k=1}^{\infty} \frac{3^k}{(k-1)!}$$

Thus, we can factor out one factor from 3^k:

$$E[X] = 3e^{-3} \sum_{k=1}^{\infty} \frac{3^{k-1}}{(k-1)!}$$

After that, we can abbreviate the summation by letting $n = k - 1$:

$$E[X] = 3e^{-3} \sum_{n=0}^{\infty} \frac{3^n}{(n)!}$$

Since the identity $E[X] = 3e^{-3}\sum_{n=0}^{\infty}\frac{3^n}{(n)!} = e^3$,

$$E[X] = 3e^{-3} \cdot e^3 = 3$$

$\therefore E[X] = 3$

To obtain the variation $Var(X) = E[X^2] - (E[X])^2$, we need to figure out $E[X^2]$. Since we know that $E[X] = 3$, $(E[X])^2 = 9$.

To expand $E[X]^2$,

$$E[X(X-1)] = E[X^2 - X] = E[X^2] - E[X]$$

Since $E[X(X-1)] = 9$, we can fill in the form:

$$9 = E[X^2] - 3$$
$$9 + 3 = E[X^2] = 12$$

Thus,

$$Var(X) = E[X^2] - E[X]^2 = 12 - 9 = 3$$

$\therefore Var(X) = 3$

2. Q2: Given two events A and B with $P(A) = 0.5, P(B) = 0.3$, and $P(A|B) = 0.6$. Consider the following:
   (a) $P(A \cap B)$
   (b) $P(B|A)$
   (c) $P(A \cup B)$

Ans: (a): To find the event $P(A \wedge B)$, we can use

$$P(A|B) = \frac{P(A \wedge B)}{P(B)}$$
$$P(A \wedge B) = P(A|B) \cdot P(B)$$
$$P(A|B) = 0.6 \cdot 0.3 = 0.18$$

(b): Using Baye's Theorem:

$$P(B|A) = P(A \wedge B)/P(A)$$
$$P(B|A) = 0.18/0.5$$
$$= 0.36$$

(c): To union two probabilities, we use the Inclusion-Exclusion Principles : $P(A) + P(B) - P(A \wedge B) = 0.5 + 0.3 - 0.18 = 0.62$

3. Q3: Suppose you have a biased coin with unknown probability of heads $\theta = P(H)$. You flip this coin 5 times and observe the sequence: H, H, T, H, T.

   (a) Derive the Maximum Likelihood Estimate (MLE) for θ, showing each step from the likelihood function.

   (b) Compute the Maximum a Posteriori (MAP) estimate for θ, assuming a prior distribution:

$$P(\theta) = \begin{cases} 2\theta, & 0 \leq \theta \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Ans:
(a) For the given sequence, we can represent it as

$$L(\theta) = P(H, H, T, H, T|\theta) = \theta \cdot \theta \cdot (1-\theta) \cdot \theta \cdot (1-\theta).$$
$$= L(\theta) = \theta^3(1-\theta)^2$$

We will use logarithmic properties to convert the given equation:

$$l(\theta) = 3\ln(\theta) + 2\ln(1-\theta)$$

Next we differentiate $l(\theta)$ with respect to θ:

$$\frac{dl}{d\theta} = \frac{d}{d\theta}(3\ln(\theta) + 2\ln(1-\theta))$$
$$\frac{dl}{d\theta} = \frac{3}{\theta} + \frac{2}{1-\theta} \cdot (-1) = \frac{3}{\theta} - \frac{2}{1-\theta}$$

We can set $\frac{dl}{d\theta} = 0$:

$$\frac{3}{\theta} - \frac{2}{1-\theta} = 0$$
$$\frac{3}{\theta} = \frac{2}{1-\theta}$$
$$= 3(1-\theta) = 2\theta$$
$$= 3 - 3\theta = 2\theta$$
$$3 = 5\theta$$
$$\theta = \frac{3}{5}$$

$\therefore$ The MLE for θ is $\frac{3}{5}$

(b) To calculate for the MAP, we need to use the prior likelihood function and distribution that was given:

$$L(\theta) = \theta^3(1-\theta)^2 \text{ for } 0 \leq \theta \leq 1$$
$$P(\theta) = 2\theta \text{ for } 0 \leq \theta \leq 1$$

We can multiply the likelihood function with the prior:

$$f(\theta) = L(\theta) \cdot P(\theta) = (\theta^3(1-0)^2) \cdot (2\theta) = 2\theta^4(1-0)^2$$

Next we can take the log of $f(\theta)$

$$ln\ f(\theta) = ln(2) + ln(\theta^4) + ln((1-\theta)^2)$$
$$= ln\ f(\theta) + 4\ ln(\theta) + 2\ ln(1-\theta)$$

We can now differentiate $ln\ f(\theta)$:

$$\frac{d}{d\theta} ln\ f(\theta) = \frac{4}{\theta} - \frac{2}{1-\theta}$$

We can set the derivative to zero and multiply:

$$\frac{4}{\theta} - \frac{2}{1-\theta} = 0$$
$$= \frac{4}{\theta} = \frac{2}{1-\theta}$$
$$= 4(1-\theta) = 2\theta$$
$$= 4 - 4\theta = 2\theta = 4 = 6\theta = \theta = \frac{4}{6}$$

$\therefore$ The MAP for the given θ and prior distribution is $\frac{4}{6}$

4. Q4: Consider a Gaussian distribution with parameters μ (mean) and σ^2 (variance). Given the dataset $X = \{1, 3, 5, 7\}$, derive the Maximum Likelihood Estimates (MLE) for both μ and σ^2. Show each mathematical step clearly.

Ans:
Using the likelihood function to calculate both the mean μ and variance σ, we can use the 1-dimensional gaussian function:

$$f(x_i|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\mu^2}} exp(-\frac{(x_i-\mu)^2}{2\sigma^2})$$

Next we can take the natural log and simplify into a sum:

$$l(\mu, \sigma^2) = ln\ L(\mu, \sigma^2) = -\frac{n}{2}ln(2\pi) - \frac{n}{2}ln(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)$$

We can now take the partial derivative of $l(\mu, \sigma^2)$:

$$\frac{\partial l}{\partial \mu} = 0 - \frac{1}{2\sigma^2}\sum_{i=1}^{n} 2(x_i = \mu)(-1) = \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i - \mu)$$

We can set the derivative to zero:

$$\frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i - \mu) = 0$$
$$\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$$

Using the given dataset X with the function: $\mu = \frac{1+3+5+7}{4} = \frac{16}{4} = 4$
To solve for variance, we need to take the partial derivative of $l(\mu, \sigma^2)$, with respect to σ^2:

$$\frac{\partial l}{\partial(\sigma^2)} = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2}\sum_{i=1}^{n}(x_i - \mu)^2$$

We can set the derivative to zero:

$$-\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2}\sum_{i=1}^{n}(x_i - \mu)^2 = 0$$

From here, can multiply the equation by $2(\sigma^2)^2$:

$$-n\sigma^2 + \sum_{i=1}^{n}(x_i - \mu)^2 = 0$$
$$= \frac{1}{n}\sum_{i-1}^{n}(x_i - \mu)^2$$

Inputting the given dataset onto the equation:

$$\sum_{i=1}^{4}(x_i - 4)^2 = (1-4)^2 + (3-4)^2 + (5-4)^2 + (7-4)^2$$
$$= \frac{20}{4} = 5$$

$\therefore$ The mean $\mu = 4$ and the variance $\sigma^2 = 5$.

5. Q5: You observe samples drawn from a Gaussian distribution with unknown mean μ and known variance $\sigma^2 = 4$. Given a Gaussian prior for the mean μ:

$$\mu \sim \mathcal{N}(0, 1)$$

derive the MAP estimate of μ after observing a single data point $x = 5$.

Ans: Using the gaussian likelihood function and prior distribution:

$$P(x|\mu) = \frac{1}{\sqrt{8\pi}} exp(-\frac{(x-\mu)^2}{8})$$
$$P(\mu) = \frac{1}{\sqrt{2\pi}} exp(-\frac{\mu^2}{2}$$

We can multiply both the likelihood and distribution together to obtain:

$$f(\mu) = exp(-\frac{(x-\mu)^2}{8} - \frac{\mu^2}{2})$$

We can then take the natural log of $f(\mu)$, and input the given x:

$$ln\ f(\mu) = -\frac{(5-\mu)^2}{8} - \frac{\mu^2}{2}$$

We can differentiate $ln\ f(\mu)$:

$$\frac{d}{d\mu} ln\ f(\mu) = -\frac{1}{8} \cdot 2(5-\mu)(-1) - \mu$$
$$= \frac{5-\mu}{4} - u$$

We can not set the derivative to zero and obtain the MAP estimate:

$$\frac{5-\mu}{4} - \mu = 0$$
$$\frac{5-\mu}{4} = \mu$$
$$(\frac{5-\mu}{4}) \cdot 4 = \mu \cdot 4$$
$$5 - \mu = 4\mu$$
$$5 = 5\mu$$
$$= 1$$

$\therefore$ The MAP estimate of μ after observing at $x = 5$ is 1.

6. Prove the variance expansion identity:

$$Var(X) = E(X^2) - [E(X)]^2$$

Ans: To prove the variance expansion identity, we will use the formal definition of the variance identity:

$$Var(X) = E[(X - E[X]^2]$$

We can expand this definition and distribute:

$$= E[X^2 - 2XE[X] + (E[X])^2$$

We can now distribute the outer expectation:

$$= E[X]^2 - 2E[XE[X]] + E[(E[X])^2]$$

Since $(E[X])^2$ is a constant, obtaining the expectation of it would only output the constant:

$$= E[X]^2 - 2E[X]E[X] + (E[X])^2 = E[X]^2 - 2(E[X])^2 + (E[X])^2$$

Finally we can subtract like terms:

$$= E[X]^2 - (E[X])^2$$

$\therefore$ It is possible to prove the variance expansion identity through its formal definition.

Part 3

7. Given a dataset represented by a random vector X, derive the mathematical expression for the variance of the projection onto a unit vector u:

$$Var(u^T X) = u^T \Sigma u$$

Document each intermediate mathematical step and define the covariance matrix Σ.

Ans: Let the covariance matrix $\Sigma =$

$$E[(X-\mu)(X-\mu)]^T$$

Assuming that our matrix is a 1x1 matrix, we can then express the mean μ, and since u is a constant in this case, it continues to be a constant after the operation:

$$E[u^T X] = u^T E[X] = u^T u$$

From here, we can apply the variance definition to $u^T X$:

$$Var(u^T X) = E[(u^T X - u^T \mu)^2]$$
$$Var(u^T X) = E[(u^T (X-\mu))^2]$$

Since we need to calculate the expectation in the variance formula, we can convert the expectation into a variable of x^2, where x represents $u^T(X-\mu)$.

We can apply a matrix transpose and have it equal to itself:

$$[u^T(X-\mu)] = [u^T(X-\mu)]^T$$
$$[u^T(X-\mu)]^T = (X-\mu)^T(u^T)^T = (X-\mu)^T u$$

Thus we can expand x^2 as $u^T(X-\mu)\cdot(X-\mu)^T u$, which itself is equivalent to having x^2 as $u^T(X-\mu)\cdot u^T(X-\mu)$, since the transpose of a scalar is equivalent.

When applying back to the variance, we can expand and convert our matrix into the covariance:

$$Var(u^T X) = E[x^2] = E[u^T(X-\mu)(X-\mu)^T u]$$
$$= u^T E[(X-\mu)(X-\mu)^T]u$$
$$= u^T \Sigma u$$

8. Compute the proportion of total variance explained by each principal component. Discuss the results and explain the significance of dimensionality reduction. Provide insights into the trade-off between explained variance and dimensionality reduction.

Ans: Since there are a total of two features, we can compute the proportion of the total variance:

$$PC1 = \frac{\lambda 1}{\lambda 1 + \lambda 2}$$

$$PC2 = \frac{\lambda 2}{\lambda 1 + \lambda 2}$$

Both calculated the two principal components respectively. Using the components that were generated from Q3, we get

$$PC1 = \frac{3.7370}{4.1772} \approx 0.8946$$
$$PC2 = \frac{0.4402}{4.1772} \approx 0.1054$$

Converting the findings to percentages, we have $PC1 \approx 89.41\%$ and $PC2 \approx 10.54\%$. Since the first principal component contains $\approx 90\%$ of the total variance, we can reduce the amount of data from two dimensions into one, while still retaining important data. As a result, this could help reduce both computation and noise when using the data. However, the trade-off between this is the loss of $\approx 11\%$ of data.